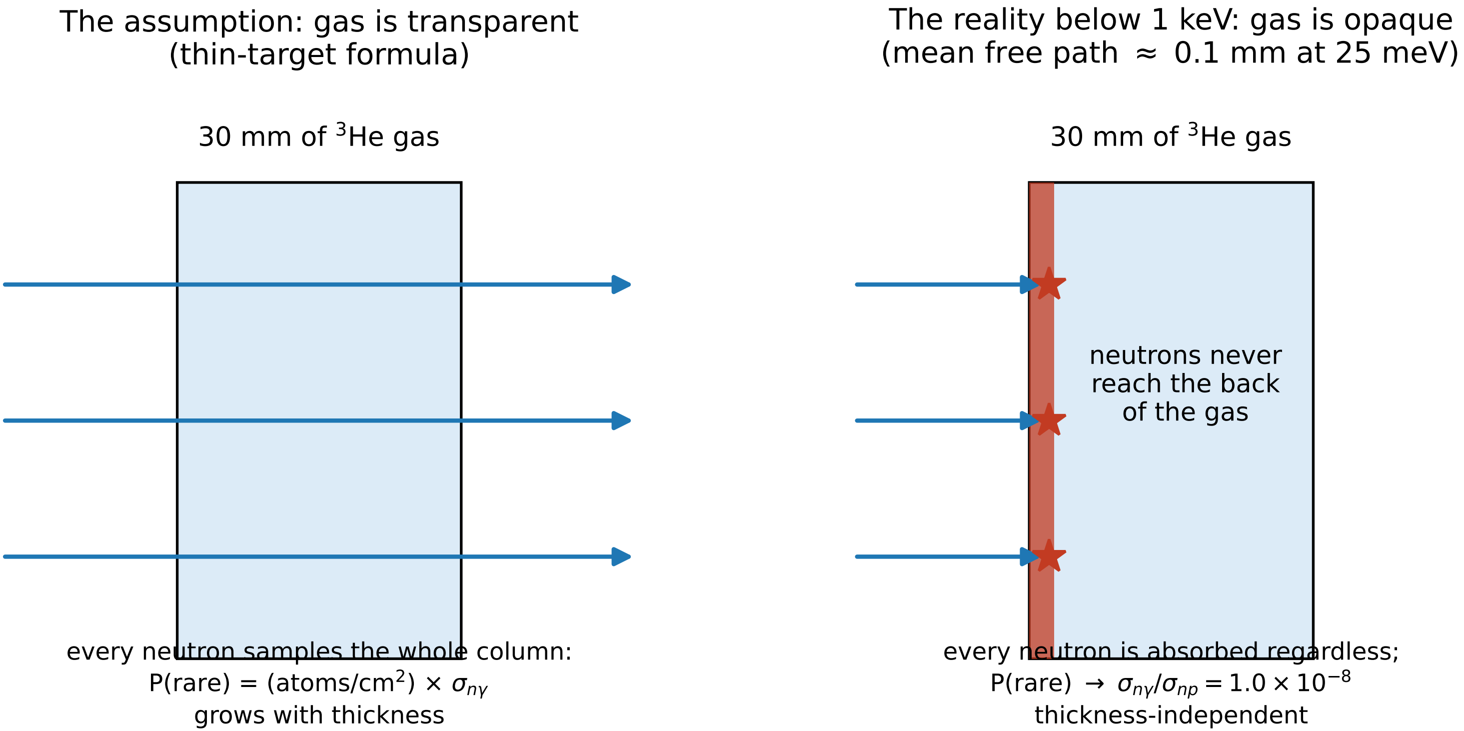
The assumption: gas is transparent
(thin-target formula)
30 mm of ^{3}He gas
every neutron samples the whole column:
P(rare) = (atoms/cm^2) × $\sigma_{n\gamma}$
grows with thickness
The reality below 1 keV: gas is opaque
(mean free path $\approx$ 0.1 mm at 25 meV)
30 mm of ^{3}He gas
neutrons never
reach the back
of the gas
every neutron is absorbed regardless;
P(rare) $\rightarrow$ $\sigma_{n\gamma}/\sigma_{np} = 1.0 \times 10^{-8}$
thickness-independent